Results: five guesses hold degree 4 at ℓ = 9 on 3 of 4 draws; the guess count stopped growing (m = 3)

Scored by `score.sh` (`score-output.txt`) against `PREREGISTRATION.md` (frozen at 37129b4253, with Amendment 1 made before any S5 branch other than d1.b0 existed) and `PREREG_SUMMAND.md`. Class: **stage diagnostic** (exact algebra on frozen draws). No `S`, no rho ratio, no speed claim, no leaderboard row. Every number is an unverified assertion `[pcc:unverified]`.

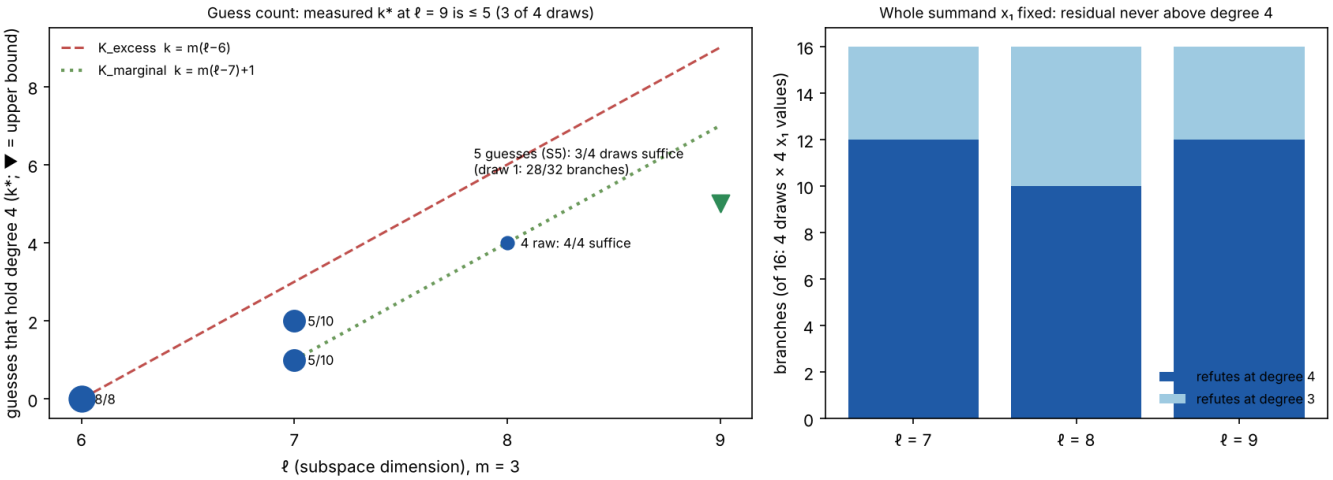


Figure: left, the guesses needed to hold the MutantXL closure at degree 4 against ℓ (m = 3). The ℓ = 6–8 points are transcribed from `kstar_recorded.json`; ▼ is this study's ℓ = 9 upper bound. The lines are the two registered models. Right, the refutation degree of the whole-summand residual, 16 branches per ℓ. Data: `results_recorded.json`, from `score-output.txt`.

Registered verdicts

- **Controls valid.**

- Gate A: the pe engine matches the committed dense records. A1 refutes at degree 4; A2 saturates unrefuted with closure rank 131,588, as committed. - The positive control P refutes at degree 4 on 4 of 4 draws. - `identity_check` is true on every one of the 128 S5 records.

- **ℓ = 9: "k* ≤ 5 at ℓ = 9 (saturation reading)".** S5 suffices on 3 of 4 draws. This

contradicts `K_marginal`, which predicted 7. My registered prediction (0.7 on "does not suffice") failed.

- **Whole summand: "mixed"** by the literal rule ("constant" needs every branch at one

degree). - Read beside it: the residual **never needs more than degree 4** at ℓ = 7, 8 or 9, and every nonzero x_1 at ℓ = 7 and ℓ = 9 refutes at exactly 4. - The degree-3 branches are $x_1 = 0$, which is degenerate (4 per ℓ), plus 2 nonzero assignments on ℓ = 8 draw 6.

The measurement

draw (3, 25, 9)	S5 branches refuting at degree 4	saturated, not refuted	verdict
1	28 / 32	4 (b28–b31)	does not suffice
4	32 / 32	0	suffices
5	32 / 32	0	suffices
7	32 / 32	0	suffices

- **Shape of a branch.** Each S5 branch has 195,709 columns at degree 4. It ran for

1,790–2,480 s at about 4.7 GB peak, with `RAYON_NUM_THREADS=2`.

- **Draw 1's failures share one corner.** They are exactly the branches with $x_3[8] = e_1[0]$

= $e_1[1] = 1$, and all four saturate at the same closure rank (188,151), whatever the two invariant guesses are. A sixth guess on that corner alone was not run.

- **Whole-summand residual.** At degree 4 the branches took 9–12 s at $\ell = 7$, 147–174 s at

$\ell = 8$ and 563–594 s at $\ell = 9$.

ℓ	refute at 3	refute at 4	nonzero x_1 only
7	4	12	12 at 4
8	6	10	10 at 4, 2 at 3
9	4	12	12 at 4

What it does and does not say

- **The recorded guess count is 0, 1–2, ≤ 4 and ≤ 5 at $\ell = 6, 7, 8$ and 9.** From $\ell = 8$ to

$\ell = 9$ it grew by at most one. `K_marginal` (m per step, 3 here) predicted three. This is one step, at m = 3, with one guess set; it is not a law.

- **Agreement with the whole-summand result.** Fixing the whole of x_1 ($k = \ell$) always lands

at degree ≤ 4. So k* ≤ ℓ holds at constant degree 4 on every cell measured. The hybrid's exponent is therefore at most about ℓ bits plus a polynomial-size closure, which is the pair table's exponent (derived).

- **No row moves.**

- Derived: at m = 3 a growth of one guess per step from k = 5 at $\ell = 9$ would give k ≈ 29 at $\ell = 33$. - The priced ECC2K-130 cell is m = 4, which is unmeasured, and its break-even is 6. - The change is that the registered extrapolation of 105 is contradicted. The guess route is not shown to beat rho, and nothing here is a speedup `[pcc:unverified]`.

- **Choice of guesses.** The guesses matter: invariant coordinates of $s = x_1 + x_2$ were

measured to beat raw ones at $\ell = 8$ `[pcc:unverified]`. A different 5-set might suffice on draw 1, or fail elsewhere.

Disclosures

- **Amendment 1 came after one S5 result.** It moved the remaining branches to a RunPod

pod after S5.d1.b0 had been recorded as refuting. The decision rule was not changed. Fail-fast was dropped, so every branch of every draw ran. Owner-approved; pod `u7l00cmnbcvnn3` (cpu3g, 64 vCPU, 256 GB, $2.56/h) ran 18:22–20:49Z on 2026-11-06, about $6.30, and was deleted (the API then returned 404). The binary's SHA-256 on the pod (`runs/pod/host.txt`) equals the local build's (`runs/binary.sha256`).

- **Interruptions, none of which changes a record.**

- The local `gateA` invocation exited 2 after both runs completed, because `run.sh` was edited while bash read it. - Two container restarts and one 2-hour task limit killed in-flight runs. Their empty attempts are kept in `runs/attempts/` and every killed run was repeated. - All are noted in `runs/queue.log`.

- **Hosts.** Gate A, the controls, S5.d1.b0 and most whole-summand runs ran in the 16 GB

container. The rest ran on the pod. Closure outcomes are exact algebra; times are indicative only.

- **Two preregistration flaws.**

- The whole-summand `h ∈ {0, 0x55, 0xAA, 0xFF}` includes the degenerate $x_1 = 0$. - For $\ell = 9$, `h = 0xFF` truncated to 9 bits leaves the top bit at 0.

Next (not run)

- **The draw-1 corner.** Add a sixth guess on $x_3[8] = e_1[0] = e_1[1] = 1$ only. That tests

whether k* = 6 there, or whether that corner needs degree 5.

- **$\ell = 10$ with the same rule.** k = 6 at S = −2 is cell (3, 28, 10). It needs a pod, at

about 236k columns.

- **m = 4.** The priced cell is m = 4. Whether the second link's invariant $e_1 + x_3$

plays the same role is the open question from `dreg_invariant_guess_more_20261029`.